Unsupervised Learning
K-means clustering

Unsupervised learning introduction

Unsupervised vs **Supervised** Learning:

- Most of this course focuses on **supervised learning** methods such as **regression** and **classification**.

- In that setting we observe both a **set of features** x_1, x_2, ..., x_p for each object, as well as a response or outcome variable Y. The goal is then to **predict Y** using $x_1, x_2, ..., x_p$.

- Here we instead focus on unsupervised learning, we where observe **only the features** $x_1, x_2, . . . , x_p$. We are not interested in prediction, because we **do not have** an associated response variable **Y**.

The Challenge of Unsupervised Learning

Unsupervised learning is **more subjective** than supervised learning, as there is no simple goal for the analysis, such as prediction of a response. But techniques for unsupervised learning are of growing importance in a number of fields:

- Subgroups of breast cancer patients grouped by their gene expression measurements.
- Groups of shoppers characterized by their browsing and purchase histories.
- Movies grouped by the ratings assigned by movie viewers.

Another advantage

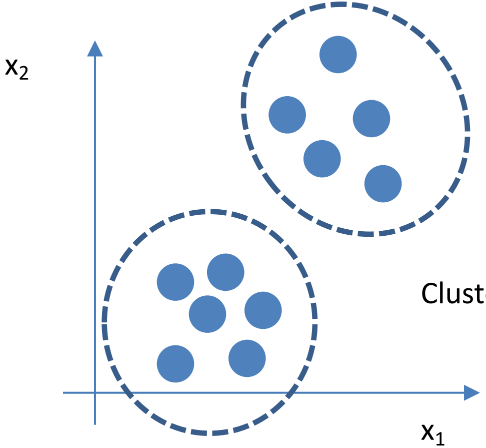
It is often **easier to obtain unlabeled data** from a lab instrument or a computer than labeled data, which can require human intervention.

For example it is difficult to automatically assess the overall sentiment of a movie review: is it favorable or not?

Clustering

- **Clustering** refers to a very broad set of techniques for finding **subgroups**, or **clusters**, in a data set.
- We seek a partition of the data into distinct groups so that the observations within each group are quite similar to each other.
- It make this concrete, we must define what it means for two or more observations to be **similar** or **different**.
- Indeed, this is often a domain-specific consideration that must be made based on knowledge of the data being studied

Unsupervised learning - Clustering



Clustering algorithm

Training set: $\{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(m)}\}$

Clustering for Market Segmentation

- Suppose we have access to a large number of measurements (e.g. *median household income occupation, distance from nearest urban area, and so forth*) for a large number of people.

- Our goal is to perform **market segmentation** by identifying subgroups of people who might be more receptive to a particular form of advertising, or more likely to purchase a particular product.

- The task of performing market segmentation amounts to clustering the people in the data set.

Two clustering methods

- In **K-means clustering**, we seek to partition the observations into a **pre-specified number of clusters**.

- In **hierarchical clustering**, we do not know in advance how many clusters we want; in fact, we end up with a tree-like visual representation of the observations, called a *dendrogram*, that allows us to view at once the clusters obtained for each possible number of clusters, from 1 to n.